# Lab 1 ticket

**Finding Similar Things — Friday, October 16**

**Lab 1 — Finding Similar Things** · Fri Oct 16 · ~20 min · Laptop · Handwrite · Initials after check-in

**Name:** _____ **Language (circle):** Octave / Python

## New syntax

| I needed to… | Command / pattern |
|---|---|
| Enter / print a vector | |
| Length or shape | |
| Distance $\|x - y\|$ | |

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**Data:** $a = (1, 2)$, $b = (2, 1)$, $c = (4, 3)$. Quiz 6 today is Week 7 matrices (**not** this lab).

**C1.0.** Enter `a`,`b`,`c`. Length of `a`: _____ **Expected:** 2.

**C1.1.** $\|a - b\|$ = _____ $\|a - c\|$ = _____
**Expected:** $\approx 1.414$ and $\approx 4.243$.

**C1.2.** NN of $a$ among $\{b, c\}$: _____ Meaning (1 sentence):

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**C1.3.** $c_{\text{bad}} = (4, 30)$. $\|a - c_{\text{bad}}\|$ = _____ NN among $\{b, c_{\text{bad}}\}$? _____
Why scale matters: _____

**Stuck on:** _____

**Check-in initials:** _____